CLASS 10 — MATHS · FORMULA SHEET

Chapter 10: **Circles** | Quick revision, one page

Tangent and secant

- A **tangent** meets the circle at exactly one point (the point of contact). A **secant** meets it at two points.
- At each point of a circle there is exactly one tangent. A circle has infinitely many tangents.
- From a point inside a circle no tangent can be drawn; from a point on it 1; from a point outside it 2.

Theorems

- The tangent at any point of a circle is **perpendicular** to the radius through the point of contact.
- The lengths of the two tangents drawn from an external point to a circle are **equal**.

Consequences used in problems

- For a point P at distance d from the centre of a circle of radius r: tangent length $= \sqrt{d^2 - r^2}$.
- If PA, PB are tangents from P with centre O: $\angle OAP = \angle OBP = 90^\circ$; $\angle AOB + \angle APB = 180^\circ$; OP bisects $\angle APB$ and $\angle AOB$.
- Concentric circles: chord of the larger circle touching the smaller has length $2\sqrt{R^2 - r^2}$.
- Tangents drawn at the two ends of a diameter are parallel.

MASTERMIND